Exercise 1B — Full Solutions

Questions 1–10 (complete)

Roots of cubic equations: symmetric functions and power sums

Colour key: green = verification check | blue = alternative method.

For a cubic $ax^3 + bx^2 + cx + d = 0$ with roots α, β, γ, Vieta gives

$$\sum \alpha = \alpha + \beta + \gamma = -\frac{b}{a}, \qquad \sum \alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \qquad \alpha\beta\gamma = -\frac{d}{a}.$$

Power-sum recurrence (multiply a root's equation $ar^3 + br^2 + cr + d = 0$ by r^n and sum over the three roots):

$$aS_{n+3} + bS_{n+2} + cS_{n+1} + dS_n = 0, \qquad S_0 = 3.$$

Seeds: $S_1 = \sum \alpha$ from Vieta, and $S_2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta$ by squaring the sum. The recurrence then delivers S_3 (at $n = 0$) and S_4 (at $n = 1$).

Question 1

Each cubic has roots α, β, γ. Find $\sum \alpha = \alpha + \beta + \gamma$ and $\alpha\beta\gamma$.

Read off $-\frac{b}{a}$ and $-\frac{d}{a}$ in each case.

(a) $x^3 + 3x^2 - 5 = 0 \quad (a = 1, b = 3, c = 0, d = -5)$:

$$\sum \alpha = -\frac{3}{1} = -3, \qquad \alpha\beta\gamma = -\frac{-5}{1} = 5.$$

(b) $2x^3 + 5x^2 - 6 = 0 \quad (a = 2, b = 5, c = 0, d = -6)$:

$$\sum \alpha = -\frac{5}{2}, \qquad \alpha\beta\gamma = -\frac{-6}{2} = 3.$$

(c) $x^3 + 7x - 9 = 0 \quad (a = 1, b = 0, c = 7, d = -9)$:

$$\sum \alpha = -\frac{0}{1} = 0, \qquad \alpha\beta\gamma = -\frac{-9}{1} = 9.$$

Check. In (c) there is no x^2 term, so the sum of roots must be 0 ✓. The constant terms are negative while $a > 0$, so each $\alpha\beta\gamma = -d/a > 0$, as found.

Question 2

$x^3 - 3x^2 + 12 = 0$ has roots α, β, γ. Find **(a)** $\sum \alpha$ and $\sum \alpha\beta$; **(b)** $\sum \alpha^2$.

Here $a = 1, b = -3, c = 0, d = 12$, so by Vieta

$$\sum \alpha = -\frac{b}{a} = 3, \qquad \sum \alpha\beta = \frac{c}{a} = 0, \qquad \alpha\beta\gamma = -\frac{d}{a} = -12.$$

(a) $\sum \alpha = 3, \qquad \sum \alpha\beta = 0.$

(b) $\sum \alpha^2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta = 3^2 - 2(0) = 9.$

Check. Numerically $\sum \alpha = 3.000$, $\sum \alpha\beta = 0.000$, $\sum \alpha^2 = 9.000$ ✓. The absent x term forces $\sum \alpha\beta = 0$.

Question 3

Roots α, β, γ. Find $S_2 = \alpha^2 + \beta^2 + \gamma^2$ and $S_{-1} = \dfrac{1}{\alpha} + \dfrac{1}{\beta} + \dfrac{1}{\gamma}$, using

$$S_2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta, \qquad S_{-1} = \tfrac{\sum \alpha\beta}{\alpha\beta\gamma}.$$

(a) $x^3 - 2x^2 + 5 = 0$: $\sum \alpha = 2,\ \sum \alpha\beta = 0,\ \alpha\beta\gamma = -5.$

$$S_2 = 2^2 - 2(0) = 4, \qquad S_{-1} = \frac{0}{-5} = 0.$$

(b) $3x^3 + 4x - 1 = 0$: $\sum \alpha = 0,\ \sum \alpha\beta = \frac{4}{3},\ \alpha\beta\gamma = \frac{1}{3}.$

$$S_2 = 0^2 - 2\left(\tfrac{4}{3}\right) = -\frac{8}{3}, \qquad S_{-1} = \frac{4/3}{1/3} = 4.$$

(c) $x^3 + 3x^2 + 5x - 7 = 0$: $\sum \alpha = -3,\ \sum \alpha\beta = 5,\ \alpha\beta\gamma = 7.$

$$S_2 = (-3)^2 - 2(5) = -1, \qquad S_{-1} = \frac{5}{7}.$$

Check. $S_2 < 0$ in (b) and (c) signals non-real roots (a real triple of numbers cannot have a negative sum of squares). For (b)/(c), solving numerically gives one real root and a complex conjugate pair, consistent with $S_2 = -\frac{8}{3}$ and $S_2 = -1$.

Question 4

$x^3 - x + 7 = 0$ has roots α, β, γ. Find $\sum \alpha$ and $\sum \alpha^2$.
Here $a = 1, b = 0, c = -1, d = 7$, so by Vieta

$$\sum \alpha = -\frac{b}{a} = 0, \qquad \sum \alpha\beta = \frac{c}{a} = -1.$$

Then

$$\sum \alpha = 0, \qquad \sum \alpha^2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta = 0 - 2(-1) = 2.$$

Check. No x^2 term, so $\sum \alpha = 0$ ✓. Numerically $\sum \alpha^2 = 2.000$ ✓; being positive, it is consistent with one real root and a complex conjugate pair.

Question 5

$2x^3 + 5x^2 + 1 = 0$ has roots α, β, γ, with $S_n = \alpha^n + \beta^n + \gamma^n$. Find S_2 and S_3.
Here $a = 2, b = 5, c = 0, d = 1$, so

$$S_1 = \sum \alpha = -\frac{5}{2}, \qquad \sum \alpha\beta = 0, \qquad \alpha\beta\gamma = -\frac{1}{2}.$$

S_2: $S_2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta = \left(-\frac{5}{2}\right)^2 - 0 = \dfrac{25}{4}.$

S_3 by the recurrence at $n = 0$: $aS_3 + bS_2 + cS_1 + dS_0 = 0$ with $S_0 = 3$ (here $a = 2, b = 5, c = 0, d = 1$):

$$2S_3 + 5S_2 + 0 \cdot S_1 + 1 \cdot S_0 = 0 \implies 2S_3 + 5\left(\tfrac{25}{4}\right) + 3 = 0 \implies 2S_3 = -\frac{137}{4} \implies S_3 = -\frac{137}{8}.$$

Alternative for S_3 — factorisation identity. Using $\sum\alpha^3 = 3\alpha\beta\gamma + \left(\sum\alpha\right)\left(\sum\alpha^2 - \sum\alpha\beta\right)$ with $\alpha\beta\gamma = -\frac{1}{2}$, $\sum\alpha = -\frac{5}{2}$, $\sum\alpha^2 = S_2 = \frac{25}{4}$, $\sum\alpha\beta = 0$:

$$S_3 = 3\left(-\tfrac{1}{2}\right) + \left(-\tfrac{5}{2}\right)\left(\tfrac{25}{4} - 0\right) = -\tfrac{3}{2} - \tfrac{125}{8} = -\tfrac{137}{8}.$$

Check. Numerically $S_2 = 6.25 = \frac{25}{4}$ ✓ and $S_3 = -17.125 = -\frac{137}{8}$ ✓.

Question 6

$x^3 + ax^2 + bx + a = 0$ has roots α, β, γ, with a, b real and positive. **(a)** Find $\sum\alpha$ and $\sum\dfrac{1}{\alpha}$ in terms of a, b. **(b)** Given $\sum\alpha = \sum\dfrac{1}{\alpha}$, does the cubic have complex roots?

The coefficients are (leading 1, then a, b, a), so by Vieta

$$\sum\alpha = -a, \qquad \sum\alpha\beta = b, \qquad \alpha\beta\gamma = -a.$$

(a) $\sum\alpha = -a$, and $\sum\dfrac{1}{\alpha} = \dfrac{\sum\alpha\beta}{\alpha\beta\gamma} = \dfrac{b}{-a} = -\dfrac{b}{a}$.

(b) The condition $\sum\alpha = \sum\dfrac{1}{\alpha}$ gives $-a = -\dfrac{b}{a}$, i.e. $a^2 = b$. Now

$$\sum\alpha^2 = \left(\sum\alpha\right)^2 - 2\sum\alpha\beta = (-a)^2 - 2b = a^2 - 2b = a^2 - 2a^2 = -a^2.$$

Since $a > 0$, $\sum\alpha^2 = -a^2 < 0$. A sum of squares of real numbers cannot be negative, so the roots are **not all real** — the cubic **has complex roots** (one real root and a conjugate pair).

Check. Take $a = 2$, so $b = a^2 = 4$: the cubic $x^3 + 2x^2 + 4x + 2 = 0$ gives $\sum\alpha = -2 = \sum\frac{1}{\alpha}$ ✓, $\sum\alpha^2 = -4 < 0$, and solving confirms one real root with a complex conjugate pair ✓.

Question 7

$x^3 - x + 3 = 0$ has roots α, β, γ. **(a)** find S_4; **(b)** evaluate $\alpha^3(\beta+\gamma) + \beta^3(\alpha+\gamma) + \gamma^3(\alpha+\beta)$.
Here $a = 1, b = 0, c = -1, d = 3$, so

$$S_1 = \sum\alpha = 0, \qquad \sum\alpha\beta = -1, \qquad \alpha\beta\gamma = -3.$$

First $S_2 = \left(\sum\alpha\right)^2 - 2\sum\alpha\beta = 0 - 2(-1) = 2$.

(a) By the recurrence at $n = 1$: $aS_4 + bS_3 + cS_2 + dS_1 = 0$ with $a = 1, b = 0, c = -1, d = 3$:

$$S_4 + 0 \cdot S_3 - S_2 + 3S_1 = 0 \implies S_4 = S_2 - 3S_1 = 2 - 3(0) = \boxed{2}.$$

(b) Since $\sum\alpha = S_1$, we have $\beta + \gamma = S_1 - \alpha$, and likewise for the others. Hence

$$\sum\alpha^3(\beta+\gamma) = \sum\alpha^3(S_1 - \alpha) = S_1\sum\alpha^3 - \sum\alpha^4 = S_1S_3 - S_4.$$

With $S_1 = 0$ this collapses to $-S_4$, so the expression $= -S_4 = \boxed{-2}$.

Check. The neat shortcut: as $S_1 = 0$, each $\beta + \gamma = -\alpha$, so the sum is $-(\alpha^4 + \beta^4 + \gamma^4) = -S_4 = -2$. Numerically the expression evaluates to -2.000 ✓.

Question 8

$2x^3 - x^2 + x - 5 = 0$ has roots α, β, γ. **(a)** Show that $2S_{n+3} - S_{n+2} + S_{n+1} - 5S_n = 0$. **(b)** Find S_{-2}.

(a) Each root r satisfies $2r^3 - r^2 + r - 5 = 0$. Multiplying by r^n,

$$2r^{n+3} - r^{n+2} + r^{n+1} - 5r^n = 0,$$

and summing over the three roots gives $2S_{n+3} - S_{n+2} + S_{n+1} - 5S_n = 0.$ ∎

(b) Here $a = 2, b = -1, c = 1, d = -5$, so by Vieta

$$\sum \alpha = \tfrac{1}{2}, \qquad \sum \alpha\beta = \tfrac{1}{2}, \qquad \alpha\beta\gamma = \tfrac{5}{2}.$$

Seeds: $S_0 = 3, \quad S_1 = \sum \alpha = \frac{1}{2}, \quad S_{-1} = \dfrac{\sum \alpha\beta}{\alpha\beta\gamma} = \dfrac{1/2}{5/2} = \frac{1}{5}.$

Apply the recurrence at $n = -2$ (its lowest term is S_{-2}): $2S_1 - S_0 + S_{-1} - 5S_{-2} = 0$, so

$$5S_{-2} = 2S_1 - S_0 + S_{-1} = 2\left(\tfrac{1}{2}\right) - 3 + \tfrac{1}{5} = -\frac{9}{5} \implies S_{-2} = -\frac{9}{25}.$$

Check. Numerically $S_{-1} = 0.200 = \frac{1}{5}$ and $S_{-2} = -0.360 = -\frac{9}{25}$ ✓.

Question 9

$px^3 + qx^2 + r = 0$ has roots α, β, γ. Find, in terms of p, q, r: **(a)** S_1; **(b)** S_2; **(c)** S_3.

Here $a = p, \; b = q, \; c = 0, \; d = r$ (note: *no* term in x), so

$$\sum \alpha = -\frac{q}{p}, \qquad \sum \alpha\beta = \frac{0}{p} = 0, \qquad \alpha\beta\gamma = -\frac{r}{p}.$$

(a) $S_1 = \sum \alpha = -\dfrac{q}{p}.$

(b) $S_2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta = \left(-\dfrac{q}{p}\right)^2 - 0 = \dfrac{q^2}{p^2}.$

(c) By the recurrence at $n = 0$: $aS_3 + bS_2 + cS_1 + dS_0 = 0$ with $S_0 = 3$ (here $a = p, b = q, c = 0, d = r$), so $pS_3 + qS_2 + 0 \cdot S_1 + 3r = 0$:

$$pS_3 + qS_2 + 3r = 0 \implies S_3 = -\frac{qS_2 + 3r}{p} = -\frac{q}{p} \cdot \frac{q^2}{p^2} - \frac{3r}{p} = -\frac{q^3}{p^3} - \frac{3r}{p} = -\frac{q^3 + 3p^2 r}{p^3}.$$

Alternative for S_3 — factorisation identity. Using $\sum \alpha^3 = 3\alpha\beta\gamma + \left(\sum \alpha\right)\left(\sum \alpha^2 - \sum \alpha\beta\right)$ with $\alpha\beta\gamma = -\frac{r}{p}$, $\sum \alpha = -\frac{q}{p}$, $\sum \alpha^2 = S_2 = \frac{q^2}{p^2}$, $\sum \alpha\beta = 0$:

$$S_3 = 3\left(-\tfrac{r}{p}\right) + \left(-\tfrac{q}{p}\right)\left(\tfrac{q^2}{p^2} - 0\right) = -\frac{3r}{p} - \frac{q^3}{p^3}.$$

Check. Test $p = 2, q = 3, r = 5$ (i.e. $2x^3 + 3x^2 + 5 = 0$): formulae give $S_1 = -\frac{3}{2}$, $S_2 = \frac{9}{4}$, $S_3 = -\frac{87}{8} = -10.875$; solving numerically reproduces $-1.500, \; 2.250, \; -10.875$ ✓.

Question 10

$x^3 + px^2 + qx + r = 0$ is such that $S_1 = 0$, $S_2 = -2$ and $S_{-1} = \dfrac{1}{5}$. Find the constants p, q, r.

For this cubic $a = 1,\ b = p,\ c = q,\ d = r$, so by Vieta

$$\sum \alpha = -p, \qquad \sum \alpha\beta = q, \qquad \alpha\beta\gamma = -r.$$

From S_1: $S_1 = \sum \alpha = -p = 0 \implies \boxed{p = 0}$.

From S_2: $S_2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta = p^2 - 2q = -2$. With $p = 0$ this gives $-2q = -2$, so $\boxed{q = 1}$.

From S_{-1}: $S_{-1} = \dfrac{\sum \alpha\beta}{\alpha\beta\gamma} = \dfrac{q}{-r} = -\dfrac{q}{r} = \dfrac{1}{5}$. With $q = 1$ this gives $-\dfrac{1}{r} = \dfrac{1}{5}$, so $\boxed{r = -5}$.

Check. With $p = 0$, $q = 1$, $r = -5$ the cubic is $x^3 + x - 5 = 0$. Solving numerically gives one real root 1.516 and a conjugate pair $-0.758 \pm 1.650\,i$, which reproduce $S_1 = 0.000$, $S_2 = -2.000$ and $S_{-1} = 0.200 = \frac{1}{5}$ ✓.